# ValeurFin — Process Flows & System Diagrams

Visual documentation of all key processes, data flows, and system architecture

v4.1 14 Diagrams April 2026 CONFIDENTIAL

## Contents


1 App Architecture Overview

2 Data Flow & State Management

3 App Boot Sequence

4 SIP Auto-Processing Flow

5 Live Price Fetch Flow

6 Add Investment Flow

7 Lot-Based XIRR Calculation

8 Date-Aware Dividend Calculation

9 Alert Generation Flow

10 Exit Signal Decision Tree

11 Wealth Score Calculation

12 Quad-Storage Fallback Chain

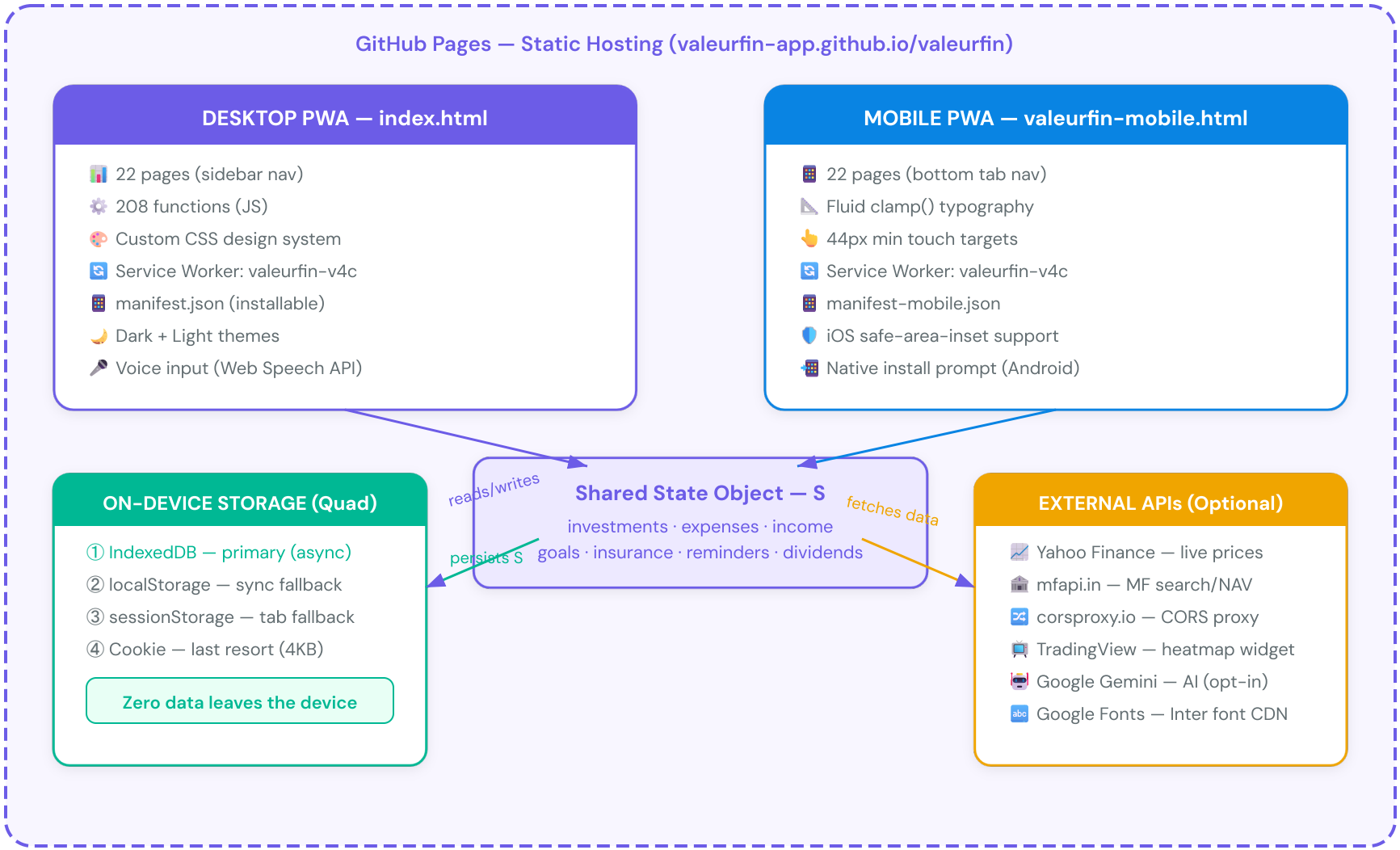
13 User Journey Map

14 ValeurBot Processing Flow

Diagram 1

# App Architecture Overview

System components, deployment, storage layers, and external API integrations



Desktop PWA Mobile PWA On-Device Storage External APIs

Diagram 2

# Data Flow — State Management

How user actions update the state object, trigger persistence, and refresh the UI

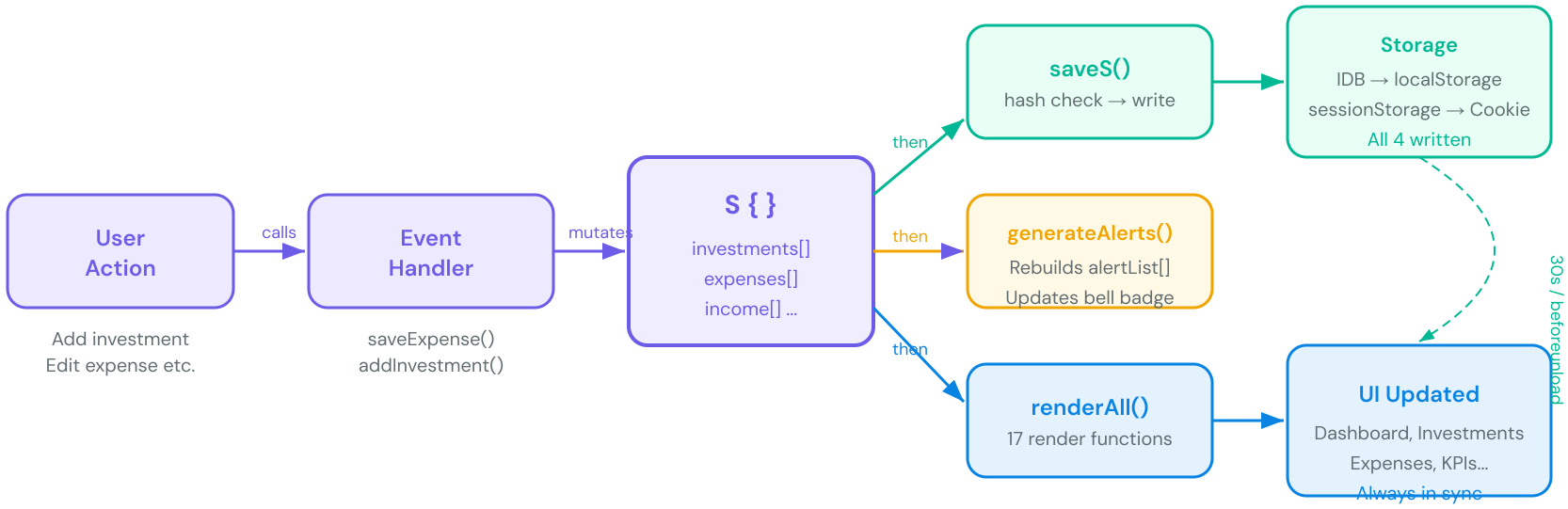


Diagram 3

# App Boot Sequence

Complete startup flow from DOMContentLoaded to app ready state

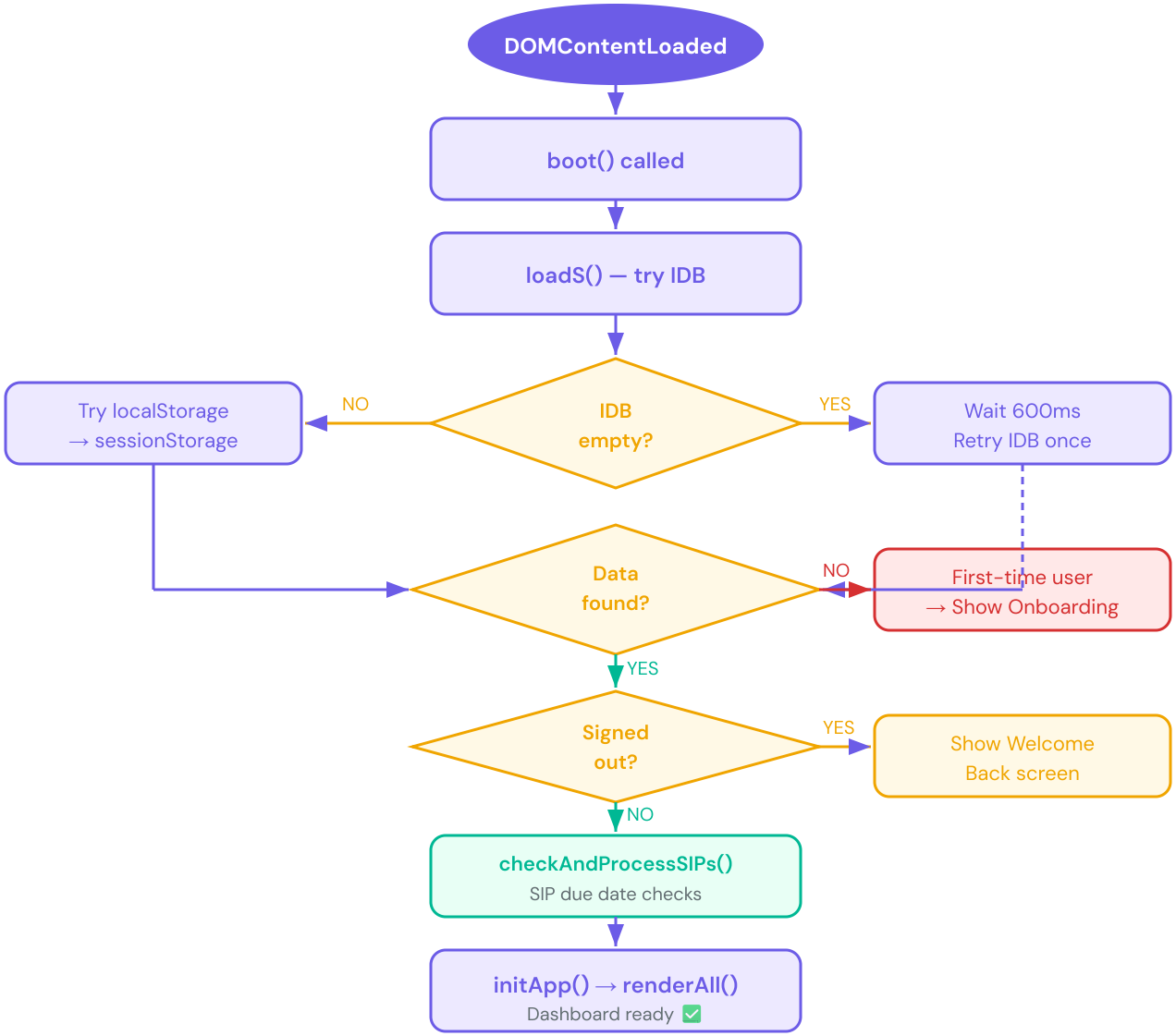


Diagram 4

## SIP Auto-Processing Flow

checkAndProcessSIPs() — runs on every app open and refresh. Zero manual clicks needed.

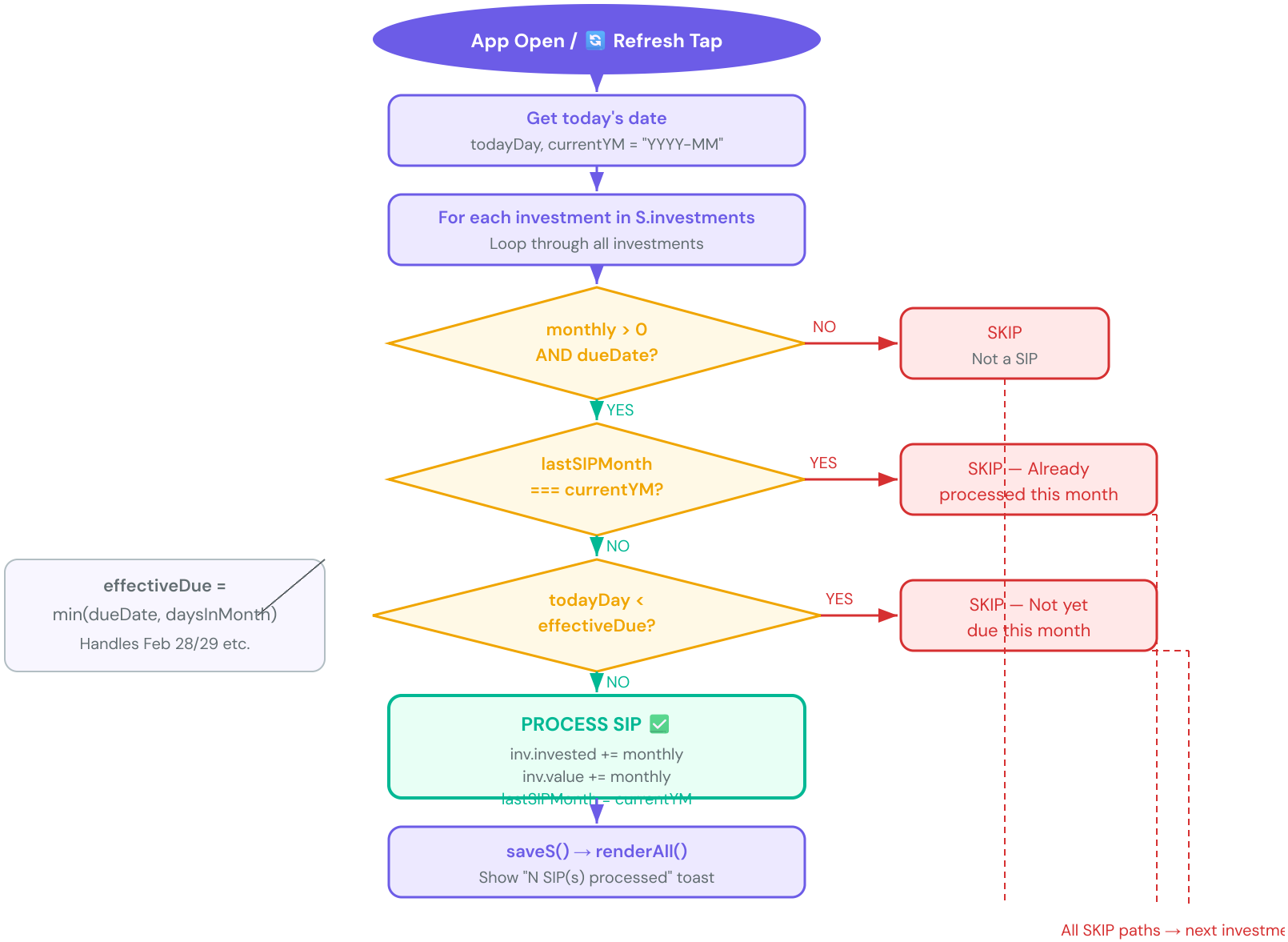

App Open / 🔄 Refresh Tap
Get today's date
todayDay, currentYM = "YYYY-MM"
For each investment in S.investments
Loop through all investments
monthly > 0
AND dueDate?
NO
SKIP
Not a SIP
YES
lastSIPMonth
=== currentYM?
YES
SKIP — Already
processed this month
NO
effectiveDue =
min(dueDate, daysInMonth)
Handles Feb 28/29 etc.
todayDay <
effectiveDue?
YES
SKIP — Not yet
due this month
NO
PROCESS SIP ✅
inv.invested += monthly
inv.value += monthly
lastSIPMonth = currentYM
saveS() → renderAll()
Show "N SIP(s) processed" toast
All SKIP paths → next investme

Diagram 5

# Live Price Fetch Flow

fetchStockPrice() — tries NSE then BSE, with CORS proxy fallback

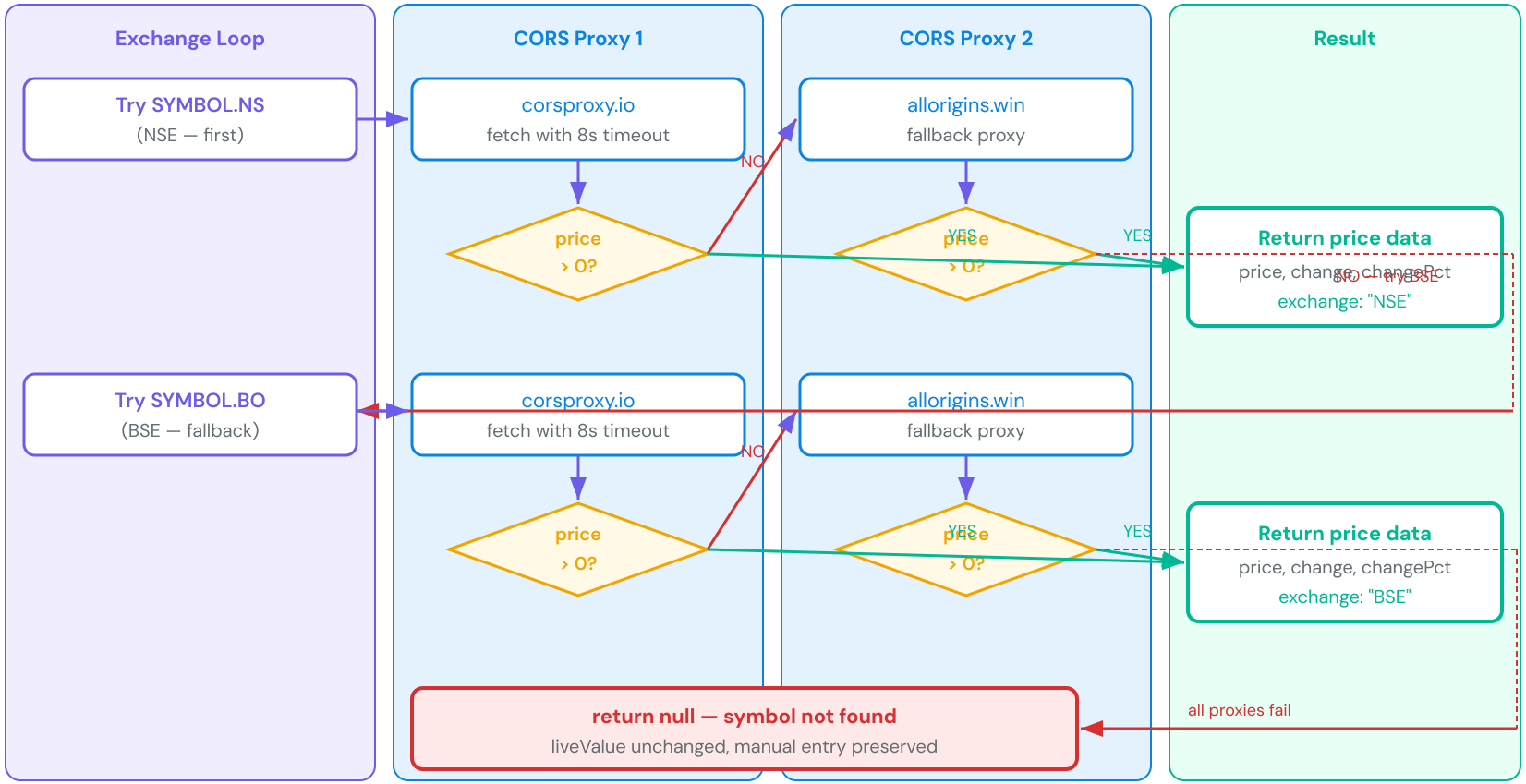
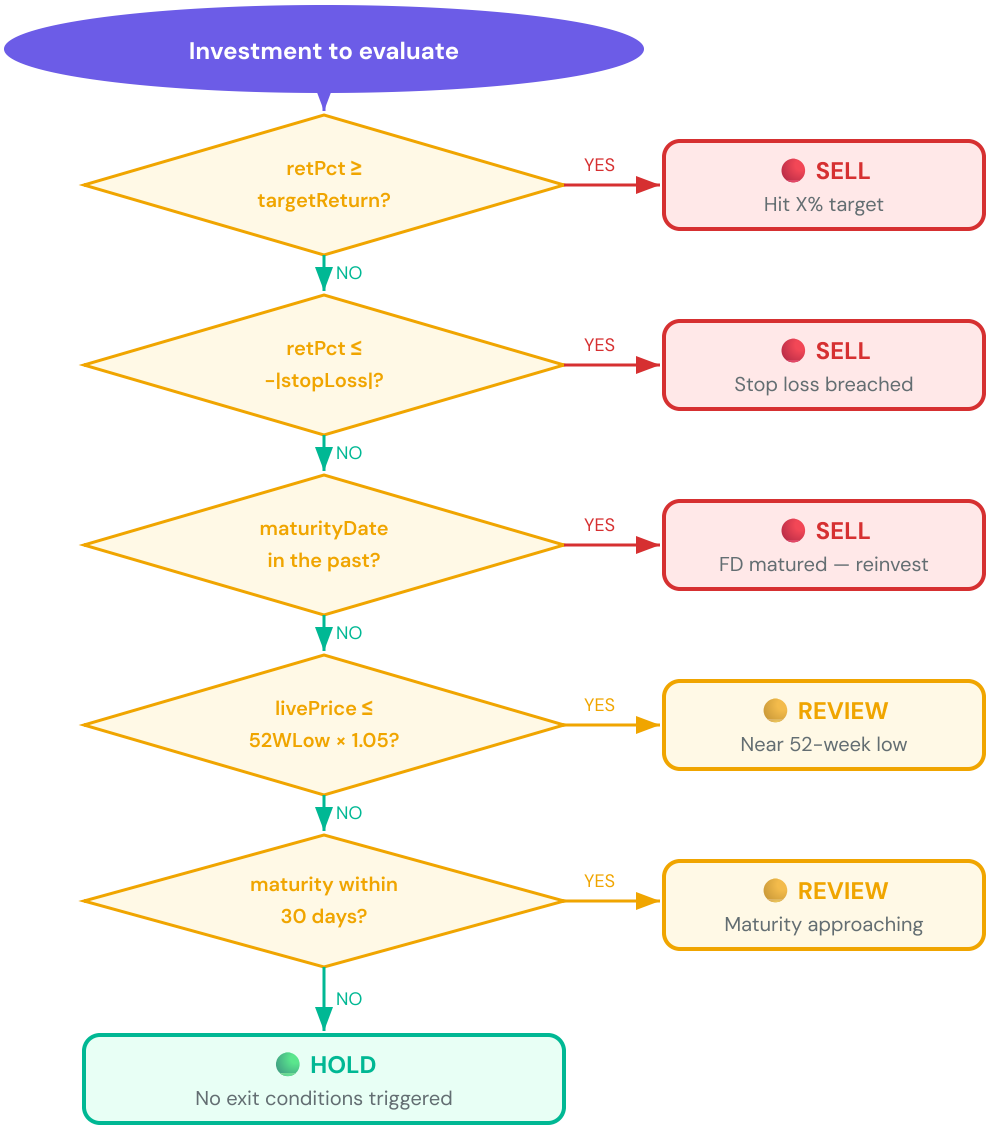


Diagram 10

# Exit Signal Decision Tree

getExitCue() — evaluated for every investment on every render. Priority order matters.




### Priority Order

Checked in sequence.
First match wins.

**SELL signals (highest priority):**

1. Target return hit
2. Stop-loss breached
3. FD/RD matured

**REVIEW signals:**

4. Near 52-week low (within 5%)
5. Maturity approaching (within 30 days)

**HOLD (default):**

None of the above triggered

User can set custom targetReturn % and stopLoss % per investment in Add/Edit form.

Diagram 11

# Wealth Score — Component Breakdown

calcWealthScore() — 7 components summed to produce a 0–100 score

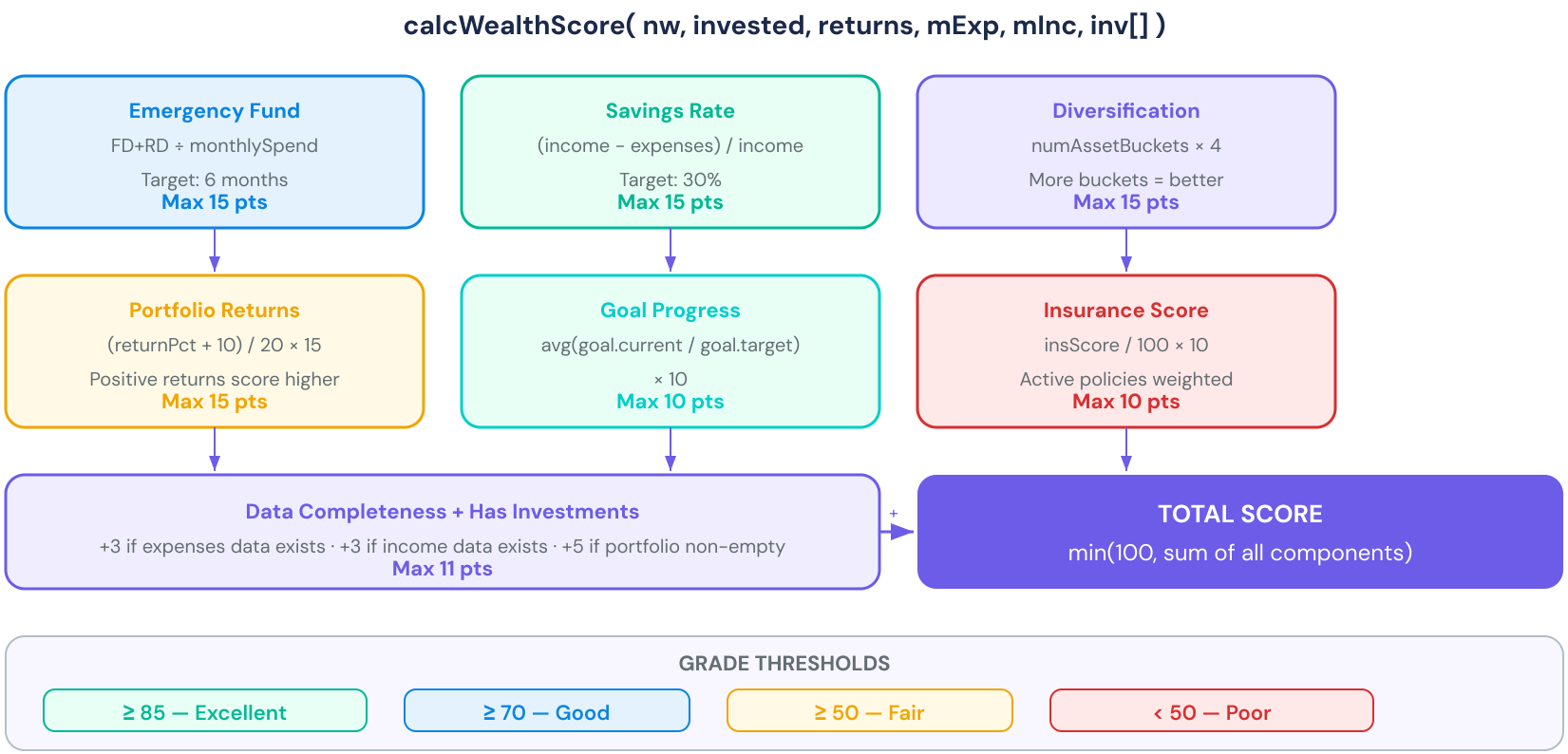


Diagram 12

# Quad-Storage Fallback Chain

loadS() and saveS() — how data is written to all 4 layers and read back

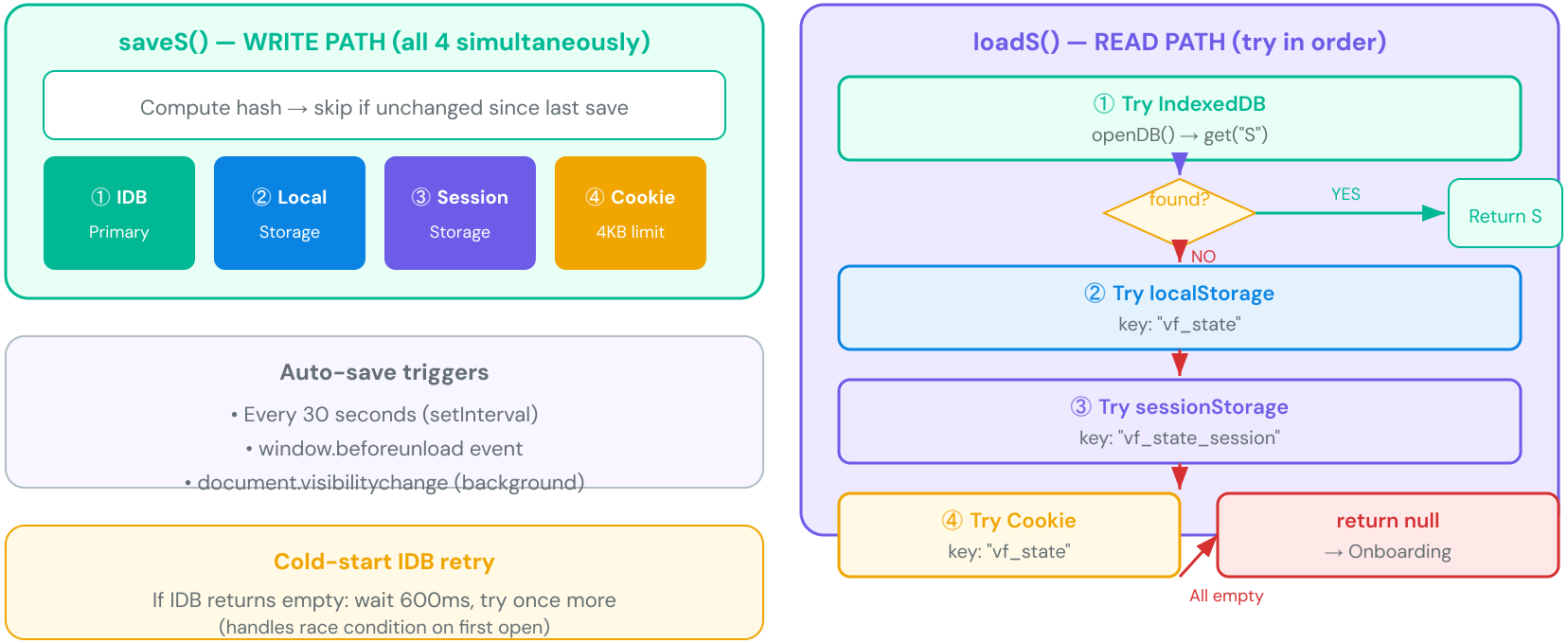


Diagram 13

# User Journey Map

First-time user vs returning user — from install to active portfolio tracking

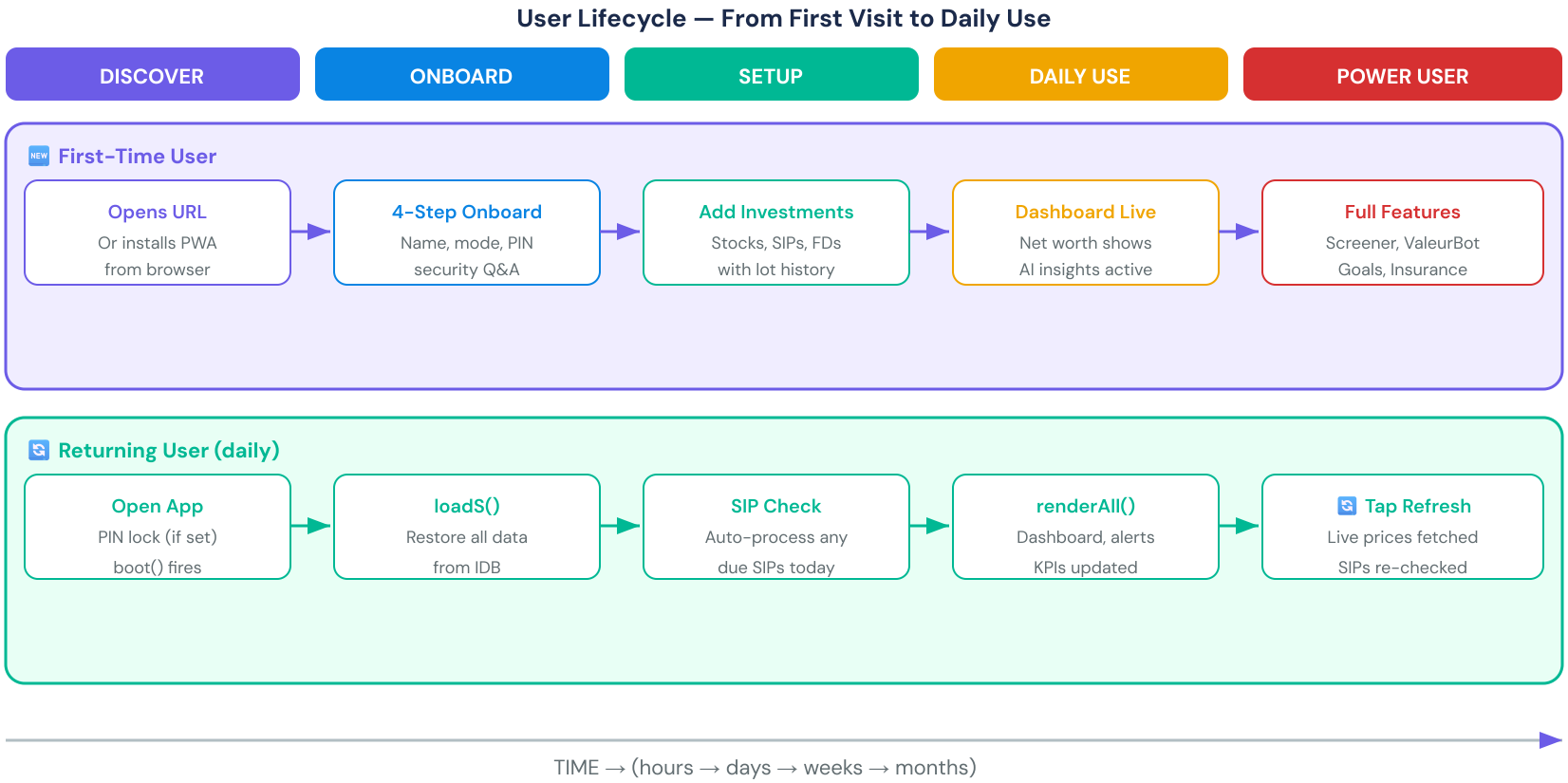


Diagram 14

# ValeurBot Processing Flow

How user queries are processed in Local mode (botAnswer) vs Gemini mode

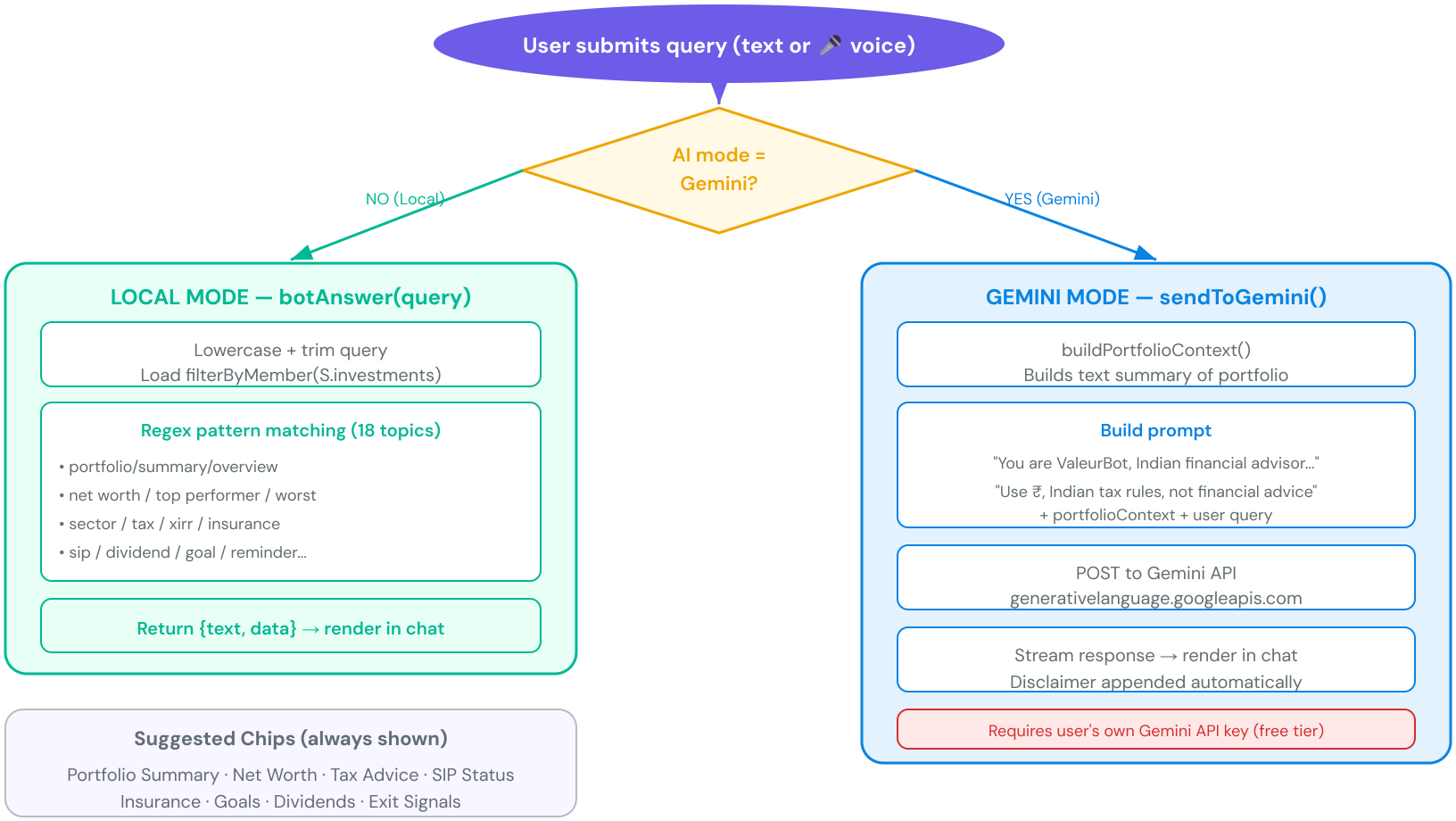
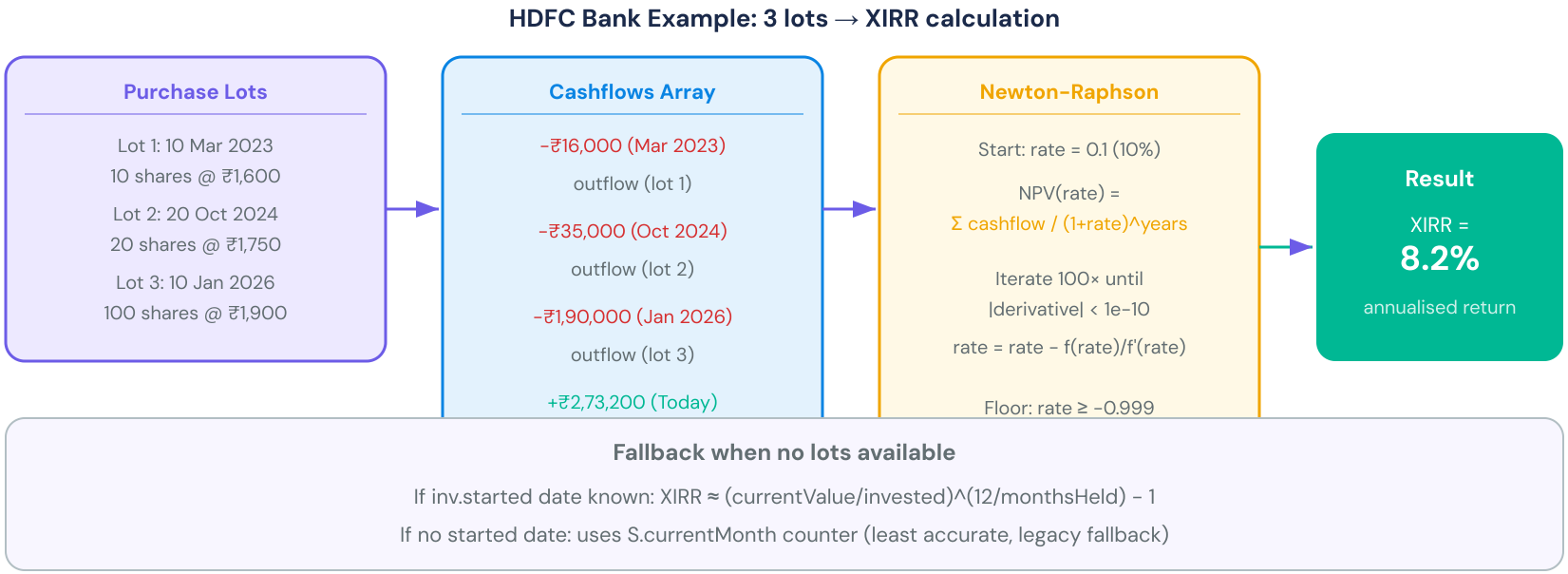


Diagram 7

# Lot-Based XIRR Calculation

calcLotsXIRR() — Newton-Raphson algorithm applied to purchase lot cashflows

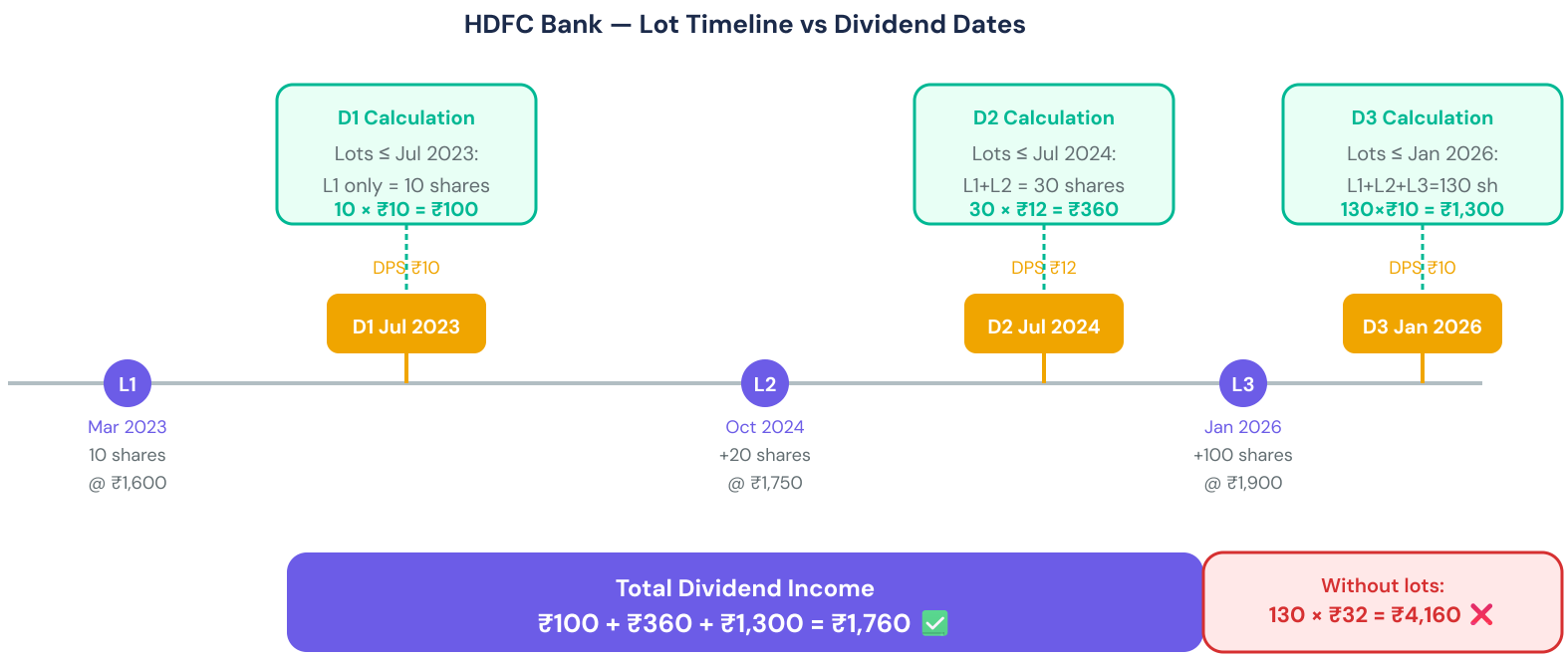



**Fallback when no lots available**

If inv.started date known: XIRR ≈ (currentValue/invested)^(12/monthsHeld) - 1

If no started date: uses S.currentMonth counter (least accurate, legacy fallback)

Diagram 8

# Date-Aware Dividend Calculation

getQtyOnDate() — how lot history determines the correct quantity for each dividend